

Formulating Capillary Surfaces in the Context of Varifolds

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- 1 Contact Angle Condition for Varifolds: Motivation and Definition
- 2 Maximum Principles for Varifolds
- 3 Open Questions

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- For any $X \in C_c^1(\mathbf{R}^{n+1}; \mathbf{R}^{n+1})$, it generates a one-parameter family $t \in [0, \infty) \mapsto f_t$ of C^1 -diffeomorphisms on $\mathbf{R}^{n+1}$, such that $f_0 = \text{id}$, with

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$$\frac{d}{dt} f_t(\cdot) = X(f_t(\cdot)).$$

- Let $\delta V[X]$ denote the first variation of the mass (area) of V with respect to X . Using the area formula, one finds that $\delta V[X]$ is given by

$$\begin{aligned} \delta V[X] &= \frac{d}{dt} \Big|_{t=0} (\mathbb{M}(f_t(V))) = \frac{d}{dt} \Big|_{t=0} \left(\int_V J_V f_t d\mathcal{H}^k \right) \\ &= \int_V \text{tr}(\nabla X \mid \text{Tan}_x V) d\mathcal{H}^k, \end{aligned}$$

where $J_V f_t$ is the tangential Jacobian of f_t with respect to V , the last integrand is also known as the **tangential divergence** of X with respect to V , usually denoted by $\text{div}_V X$.

General Varifolds I: Introduction

The theory of (general) varifolds ¹ generalizes the previous argument to the measure-theoretic setting (non-smooth case).

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- For $1 \leq k \leq n + 1$, a k -varifold V on a given subset $U \subset \mathbf{R}^{n+1}$ is a positive Radon measure on $G_k(U)$, where $G_k(U) := U \times G(n + 1, k)$, for $G(n + 1, k)$ the Grassmannian of the un-oriented k -planes of $\mathbf{R}^{n+1}$.

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- The weight measure of the general varifold V , denoted by $\|V\|$, is the positive Radon measure on U , defined as

$$\|V\|(A) = V(G_k(A)), \text{ for any Borel set } A \subset U.$$

Correspondingly, the mass $\mathbb{M}(V)$ of V is defined by

$$\mathbb{M}(V) = \|V\|(U) = V(G_k(U)).$$

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General Varifolds I: Introduction

- For any C^1 -diffeomorphism f on $\mathbf{R}^{n+1}$, the **image varifold** $f_{\#}V$ is the k -varifold on $f(U)$, given by: for any $\varphi \in C_c(G_k(f(U)))$,

$$\int_{G_k(f(U))} \varphi(y, T) df_{\#}V(y, T) = \int_{G_k(U)} J_S f(x) \varphi(f(x), df_x(S)) dV(x, S),$$

where $J_S f(x)$ is the Jacobian of f relative to the k -plane S , i.e.,

$$J_S f(x) = \sqrt{\det \left((df_x)_{|S}^* \circ ((df_x)_{|S}) \right)}.$$

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Clearly,

$$\mathbb{M}(f_{\#}V) = \int_{G_k(U)} J_S f(x) dV(x, S).$$

General Varifolds I: Introduction

- For any $X \in C_c^1(\mathbf{R}^{n+1}; \mathbf{R}^{n+1})$ and the corresponding one-parameter family $t \in [0, \infty) \mapsto f_t$, the well-known first variation formula for mass of k -varifold:

$$\begin{aligned}\delta V[X] &= \frac{d}{dt} \Big|_{t=0} \mathbb{M}((f_t)_\# V) = \int_{G_k(U)} \operatorname{tr}(\nabla X(x) \mid S) dV(x, S) \\ &= \int_{G_k(U)} \operatorname{div}_S X(x) dV(x, S).\end{aligned}$$

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- The first variation formula of varifolds is consistent with that of C^2 -surfaces, as given a k -dim, C^2 -surface $M \subset U \subset \mathbf{R}^{n+1}$, we can relate it naturally to the k -varifold

$$V = \mathcal{H}^k \llcorner M \otimes \delta_{T_x M}.$$

Consequently,

$$\delta V[X] = \int_{G_k(U)} \text{div}_S X(x) dV(x, S) = \int_M \text{div}_M X(x) d\mathcal{H}^k.$$

General Varifolds II: Surfaces without Boundary

- Given an open set $U \subset \mathbf{R}^{n+1}$, and a k -dim, compact, C^2 -surface $V \subseteq U$ without boundary, the first variation formula of $\mathbb{M}(V)$ with respect to $X \in C_c^1(U; \mathbf{R}^{n+1})$ is given by

$$\begin{aligned}\delta V[X] &= \int_V \operatorname{div}_V X \, d\mathcal{H}^k = \int_V \operatorname{div}_V (X^T + X^\perp) \, d\mathcal{H}^k \\ &= - \int_V \langle X, \vec{H} \rangle \, d\mathcal{H}^k,\end{aligned}$$

where (at any $x \in V$, let $\{\tau_i(x)\}_{i=1,\dots,k}$ be an orthonormal basis of $T_x V$)

$$\begin{aligned}\operatorname{div}_V (X^\perp) &= \sum_{i=1}^k \langle \nabla_{\tau_i} (\langle X, \nu \rangle \nu), \tau_i \rangle \\ &= \left\langle X, \sum_{i=1}^k \langle \nabla_{\tau_i} \nu, \tau_i \rangle \nu \right\rangle =: - \langle X, \vec{H} \rangle.\end{aligned}$$

General Varifolds II: Surfaces without Boundary

- A natural generalization to k -varifolds:

Let $\mathfrak{X}_c(U)$ denote the set of C^1 -vector fields on U with compact support in U .

Given an open set $U \subset \mathbf{R}^{n+1}$ and a k -varifold V on U , V is said to have generalized mean curvature with respect to $\mathfrak{X}_c(U)$, if there exists a $\|V\|$ -measurable vector field $\vec{H} \in L^1(M, \|V\|)$, such that the following formula holds:

$$\delta V[X] = \int_{G_k(U)} \operatorname{div}_S X(x) dV(x, S) = - \int_U \langle X, \vec{H} \rangle d\|V\|.$$

This is the weak formulation of k -surface that is contained in U , without boundary.

General Varifolds II: Surfaces without Boundary

- Alternatively, if: **firstly**, V has bounded first variation with respect to $\mathfrak{X}_c(U)$, in the sense that

$$\sup\{|\delta V[X]| : X \in \mathfrak{X}_c(U), \max|X| \leq 1\} < +\infty,$$

then the Riesz representation theorem yields a positive Radon measure $\|\delta V\|$ on U and a $\|\delta V\|$ -measurable vector field ν with $|\nu| = 1$ $\|\delta V\|$ -a.e. in U , such that

$$\delta V[X] = - \int_U \langle X, \nu \rangle d\|\delta V\|.$$

General Varifolds II: Surfaces without Boundary

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Secondly, $\|\delta V\| \ll \|V\|$.

- In this case, the V has generalized mean curvature with respect to $\mathfrak{X}_c(U)$, given by

$$\vec{H} = D_{\|V\|} \|\delta V\| \cdot \nu.$$

Surfaces with Free Boundary

- Given an open set $U \in \mathbf{R}^{n+1}$ with smooth boundary ∂U .
A compact, k -dim, C^2 -surface $V \in \overline{U}$ is said to have free boundary at ∂U if $\partial V \subset \partial U$, and V meets ∂U perpendicularly.
This means, if we denote by Ω the enclosing region of V with ∂U , and by ν, ν_U the inwards pointing (with respect to Ω) unit normal along $V, \partial\Omega \cap \partial U$, respectively, and μ the inwards pointing unit co-normal along ∂V in V , then

$$\mu = \nu_U \text{ along } \partial V.$$

Surfaces with Free Boundary

- Any such free boundary surface V satisfies the following distribution structure:

Let $\mathfrak{X}_t(U)$ denote the set of C^1 -vector fields defined on $\overline{U}$ which are tangent to ∂U .

The first variation of the mass of V with respect to any $X \in \mathfrak{X}_t(U)$ is then given by

$$\delta V[X] = \int_V \operatorname{div}_V X \, d\mathcal{H}^k = - \int_V \langle X, \vec{H} \rangle \, d\mathcal{H}^k,$$

since (recall that $\mu = \nu_N$ along ∂V)

$$\int_V \operatorname{div}_V (X^T) \, d\mathcal{H}^k = \int_{\partial V} \langle X, -\mu \rangle \, d\mathcal{H}^{k-1} = 0.$$

Varifolds with Free Boundary

Definition (Varifolds with free boundary)

A k -varifold V on U is said to *with free boundary at ∂U* , if there exists a $\|V\|$ -measurable vector field $\vec{\mathbf{H}} \in L^1(\overline{U}, \|V\|)$ such that for any $X \in \mathfrak{X}_t(U)$, the distributional structure

$$\delta V[X] = - \int_{\overline{U}} \langle X, \vec{\mathbf{H}} \rangle d\|V\|$$

holds.

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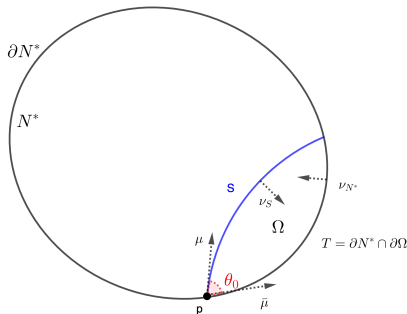
holds.

- Note that in this definition, $\|V\|(\partial U)$ is allowed to be non-negligible, which is significantly different from the C^2 -surface case.
- Since we only test δV with X that is tangent to ∂U , to make the choice of $\vec{\mathbf{H}}$ unique, if not otherwise stated, we always assume that $\vec{\mathbf{H}}$ is tangent to ∂U for $\|V\|$ -a.e. $x \in \partial U$.

Surfaces with Capillary Boundary

- A compact, n -**dim**, C^2 -hypersurface $V \in \overline{U}$ is said to **have fixed contact angle** θ (or capillary boundary) at ∂U if $\partial V \subset \partial U$, and V meets ∂U with a fixed contact angle $\theta \in (0, \pi)$. That is,

$$\mu = \cos \theta \bar{\mu} + \sin \theta \nu_N \text{ along } \partial V.$$



Surfaces with Capillary Boundary

- Any such capillary surface V satisfies the following distribution structure:

The first variation of the mass of V with respect to any $X \in \mathfrak{X}_t(U)$ is given by

$$\delta V[X] = \int_V \operatorname{div}_V X \, d\mathcal{H}^n = \int_{\partial V} \langle X, -\mu \rangle \, d\mathcal{H}^{n-1} - \int_V \langle X, \vec{\mathbf{H}} \rangle \, d\mathcal{H}^n,$$

meanwhile, one has

$$\delta W[X] = \int_W \operatorname{div}_W X \, d\mathcal{H}^n = \int_W \operatorname{div}_{\partial U} X \, d\mathcal{H}^n = \int_{\partial W = \partial V} \langle X, -\bar{\mu} \rangle \, d\mathcal{H}^n.$$

Thus from the capillary condition $\mu = \cos \theta \bar{\mu} + \sin \theta \nu_N$ along ∂V , we deduce

$$(\delta V - \cos \theta \delta W)[X] = - \int_V \langle X, \vec{\mathbf{H}} \rangle \, d\mathcal{H}^n.$$

Varifolds with Fixed Contact Angle

The first attempt in this direction:

Definition (Contact Angle Condition, Takashi Kagaya and Yoshihiro Tonegawa ^{a)})

^{a)}A fixed contact angle condition for varifolds, Hiroshima Math. J. **47** (2017), no.2, 139–153.

Given an n -varifold V on U with finite mass and $\|V\|(\partial U) = 0$, a $\mathcal{H}^n$ -measurable set $T \subset \partial U$, and $\theta \in (0, \pi)$. V is said to *have a fixed contact angle θ with ∂U at the boundary of T* if the following distributional structure holds:

$$\delta V[X] - \cos \theta \int_T \operatorname{div}_{\partial U} X d\mathcal{H}^n = - \int_U \langle X, \vec{\mathbf{H}} \rangle d\|V\|,$$

where $X \in \mathfrak{X}_t(U)$ and $\vec{\mathbf{H}} \in L^1(U, \|V\|)$ is a $\|V\|$ -measurable vector field.

Varifolds with Fixed Contact Angle

A recent generalization:

Definition (Contact Angle Condition, Luigi De Masi and Guido De Philippis ^a)

^a*Min-max construction of minimal surfaces with a fixed angle at the boundary*, (2021), arXiv: 2111.09913

Given $\theta \in [\frac{\pi}{2}, \pi)$, an n -varifold V on $\overline{U}$, an n -varifold W on ∂U . The pair of varifolds (V, W) is said to satisfy the contact angle condition θ if there exists a $\|V\|$ -measurable vector field $\vec{H} \in L^1(\overline{U}, \|V\|)$, such that for any $X \in \mathfrak{X}_t(U)$, there holds

$$(\delta V - \cos \theta \delta W)[X] = - \int_{\overline{U}} \langle X, \vec{H} \rangle d\|V\|.$$

Varifolds with Fixed Contact Angle

- Compared to the definition by Kagaya-Tonegawa, the condition on W is relaxed. Since W is now a positive Radon measure, we are free to take $V - \cos \theta W$ ($\theta \in [\frac{\pi}{2}, \pi)$), which gives a new varifold.

Varifolds with Fixed Contact Angle

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- The nice thing is that $V - \cos \theta W$ has free boundary at ∂U .

Lemma (Luigi De Masi ^a)

^a*Existence and properties of minimal surfaces and varifolds with contact angle conditions*, (2022), Thesis (Ph.D.)–SISSA

Given n -varifold V on $\overline{U}$ and n -varifold W on ∂U , if (V, W) satisfies the contact angle condition θ , then $V - \cos \theta W$ has free boundary at ∂U , with generalized mean curvature $\hat{\mathbf{H}}$ satisfying

$$\int_A \vec{\mathbf{H}} d\|V\| = \int_A \hat{\mathbf{H}} d(\|V\| - \cos \theta \|W\|), \text{ for any Borel set } A \subset \overline{U}.$$

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Interior Maximum Principle

- Let $N^* \subset \mathbf{R}^{n+1}$ be an open domain with smooth boundary ∂N^* .
- Let $\mathfrak{X}(N^*)$ denote the set of compactly supported C^1 vector field on $\overline{N^*}$ such that $\langle X, \nu_{N^*} \rangle \geq 0$ along ∂N^* .
A k -varifold V in N^* is called *minimizes area to first order in N^** , if $\delta V[X] \geq 0$ for any $X \in \mathfrak{X}(N^*)$.
- N^* is called *strongly k -convex* at a point $p \in \partial N^*$ if

$$\kappa_1 + \dots + \kappa_k > 0,$$

where $\{\kappa_i\}_{i=1}^n$ are the principal curvatures of $\partial N^* \subset \mathbf{R}^{n+1}$ at p , indexed in the increasing order.

Interior Maximum Principle

Theorem (Interior Maximum Principle, Brian White ^a)

^a*The maximum principle for minimal varieties of arbitrary codimension*, Comm. Anal. Geom. **18** (2010), no.3, 421–432.

Let p be a point in ∂N^ at which N^* is strongly k -convex, then p is not contained in the support of any k -dimensional varifold V in N^* that minimizes area to first order in N^* .*

¹*The barrier principle for minimal submanifolds of arbitrary codimension*, Ann. Global Anal. Geom. **24** (2003), no.3, 261–267.

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- By support of V we mean, $\text{spt}\|V\|$.
- In case V is a smooth minimal submanifold, the fact that V cannot contain p was proved by Luquésio Jorge and Friedrich Tomi ¹.

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- By support of V we mean, $\text{spt}\|V\|$.
- In case V is a smooth minimal submanifold, the fact that V cannot contain p was proved by Luquésio Jorge and Friedrich Tomi ¹.
- In addition, in the codimension-1 case, this follows from the classical Hopf's lemma.

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Maximum Principle for Free Boundary Varifolds

- Let $N^* \subset \mathbf{R}^{n+1}$ be an open domain with smooth boundary ∂N^* .
- Let $\mathfrak{X}_t(N^*)$ denote the set of compactly supported C^1 -vector field on $\overline{N^*}$ such that $\langle X, \nu_{N^*} \rangle = 0$ along ∂N^* .
- Let $N \subset N^*$ be a compact, connected closed set whose relative boundary in N^* , say $S = \partial_{\text{rel}} N$, is a properly embedded smooth hypersurface with free boundary at ∂N^* .
- A k -varifold V in N is said to be *stationary with free boundary at ∂N^** if $\delta V[X] = 0$ for any $X \in \mathfrak{X}_t(N^*)$.

Maximum Principle for Free Boundary Varifolds

The free boundary version of Brian White's result.

Theorem (Boundary Maximum Principle for Free Boundary Varifolds, Martin Man-Chun Li and Xin Zhou ^a)

^aA maximum principle for free boundary minimal varieties of arbitrary codimension, Comm. Anal. Geom. **29** (2021), no.6, 1509–1521.

Let $N \subset N^$ be a compact, connected closed set whose relative boundary in N^* , say $S = \partial_{\text{rel}} N$, is a properly embedded smooth hypersurface with free boundary at ∂N^* .*

If S is strongly k -convex at a point $p \in \partial S$, then p is not contained in the support of any k -dimensional stationary varifold V with free boundary at ∂N^ , which is supported in N .*

Strategy of Proof I: Interior Maximum Principle

Brian White's proof-crucial lemma:

Lemma

Let p be a point in ∂N^ at which $0 < c_p < \kappa_1 + \dots + \kappa_k$. Then there exists a compactly supported C^∞ -vector field X with $X(p)$ a nonzero (scalar multiple) normal to ∂N^* , such that*

$$\langle X, \nu_{N^*} \rangle \geq 0 \text{ along } \partial N^*, \text{ and } \delta V[X] \leq -c_p \int |X| d\|V\|$$

for any k -varifold V in N^ .*

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for any k -varifold V in N^ .*

Suppose for a while that this lemma is true, then provided that V minimizes area to first order in N^* , we immediately deduce

$$0 \leq \delta V[X] \leq -c_p \int |X| d\|V\| \leq 0,$$

implying that $|X|$ vanishes $\|V\|$ -a.e., and hence $\text{spt}\|V\|$ cannot contain any point where $X \neq 0$, for example p .

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- Since Σ is smooth, from the existence of tubular neighborhood $\mathcal{U}$, one sees that s is smooth near Σ .

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- For any small number $|t| \ll 1$, let Σ_t denote the level-set of $s(x)$ with value t . Clearly $\{\Sigma_t\}_{|t| \ll 1}$ forms a smooth foliation of the tubular neighborhood $\mathcal{U}$.

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- For any small number $|t| \ll 1$, let Σ_t denote the level-set of $s(x)$ with value t . Clearly $\{\Sigma_t\}_{|t| < 1}$ forms a smooth foliation of the tubular neighborhood $\mathcal{U}$.
- By setting $\nu(q) := \nabla s(q)$, which is exactly the inwards pointing unit normal at every $q \in \bigcup_{t: |t| < 1} \Sigma_t$, we obtain a smooth vector field near Σ .

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- (smooth cut-off function) For any $0 < \epsilon$, define

$$\phi(t) = \begin{cases} e^{\frac{1}{t-\epsilon}}, & 0 \leq t < \epsilon, \\ 0, & t \geq \epsilon. \end{cases}$$

$$\phi'(t) = -\frac{1}{(t-\epsilon)^2} e^{\frac{1}{t-\epsilon}}.$$

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- If we look at $\phi(s(\cdot))$, then we can study the behavior near $\Sigma = \Sigma_0$, to only see things near p , we modify Σ . Precisely, we replace Σ ($s(\cdot)$ and ν) in the above argument by

$$\tilde{\Sigma} := \{q \in \mathbf{R}^{n+1} \setminus N^* : \text{dist}(q, \partial N^*) = \text{dist}^4(q, p)\},$$

a hypersurface in $\mathbf{R}^{n+1}$ that touches N^* from outside at p , up to second order.

Strategy of Proof I: Interior Maximum Principle

- Define $X(q) = \phi(s(q))\nu(q)$, which is a smooth vector field near p , with $X(p) = e^{-\frac{1}{\epsilon}}\nu(p)$ and $\langle X, \nu_{N^*} \rangle \geq 0$ along ∂N^* .

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- The strongly k -convexity of ∂N^* at p will then force

$$\delta V[X] = \int \operatorname{tr}(\nabla X \mid S) dV(x, S) \leq -c_p \int |X| d\|V\|,$$

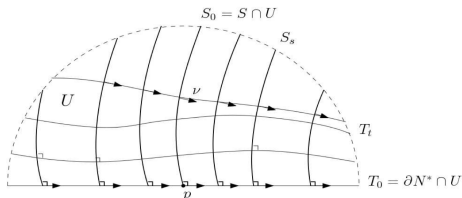
for any k -varifold that is supported in N^* .

Strategy of Proof II: Boundary Maximum Principle for Varifolds with Free Boundary

The key point of Li-Zhou's argument is to find a free boundary version of the local smooth foliation for the strictly k -mean convexity point $p \in \partial S$.

Lemma (Local Foliation near $p \in \partial S \subset \partial N^*$)

There exists a constant $\delta > 0$, a neighborhood $U \subset N^$ containing p and foliations $\{S_s\}$ with $s \in (-\delta, \delta)$, of U such that $S_0 = S \cap U$, and each leaf S_s is a smooth hypersurface with free boundary at ∂N^* .*



Strategy of Proof II: Boundary Maximum Principle for Varifolds with Free Boundary

- To construct $\{S_s\}$, consider the local Fermi coordinate system $(x_1, \dots, x_{n+1})$ of $\mathbf{R}^{n+1}$ centered at p , relative to the hypersurface ∂N^* , such that $x_1 = \text{dist}(\cdot, \partial N^*)$.
- Since ∂S is a hypersurface in ∂N^* , we may further assume that $(x_2, \dots, x_{n+1})$ is a local Fermi coordinate system of ∂N^* , that is, x_{n+1} is the signed distance function on ∂N^* from ∂S .

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- Because S has free boundary at ∂N^* , we can write S locally in this Fermi coordinate as graph $x_{n+1} = f(x_1, \dots, x_n)$, with $f \equiv 0$ on $\{x_1 = 0\}$ and $\frac{\partial f}{\partial x_1} \equiv 0$ on $\{x_1 = 0\}$ (due to free boundary condition).

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- Write $f_s(x_1, \dots, x_n) = s + f(x_1, \dots, x_n)$, the translated graphs $(x_1, \dots, x_n, f_s(x_1, \dots, x_n))$ then gives the desired foliation $\{S_s\}$. One could show that each leaf S_s has free boundary at ∂N^* .

Strategy of Proof II: Boundary Maximum Principle for Varifolds with Free Boundary

- For every $q \in S_s$, let $\nu(q)$ be a unit normal to S_s , by a continuous choice of $\nu(q)$ (such that if $q \in S_0 = S$, $\nu(q)$ points inwards N), we obtain a smooth vector field ν in N^* , locally near P .
- Finally, the test vector field X is defined as

$$X(q) = \phi(s(q))\nu(q),$$

which is clearly tangent to ∂N^* .

A lengthy computation then shows that, if $p \in \text{spt}\|V\|$ for some k -stationary varifold that is supported in N with free boundary at ∂N^* , then

$$0 = \delta V[X] < 0,$$

a contradiction.

Main Result

Question: Can the previous maximum principles be generalized to hold for varifolds with fixed contact angle θ ?

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No and **Yes**.

Not valid for every (V, W) that satisfies fixed contact angle condition θ .

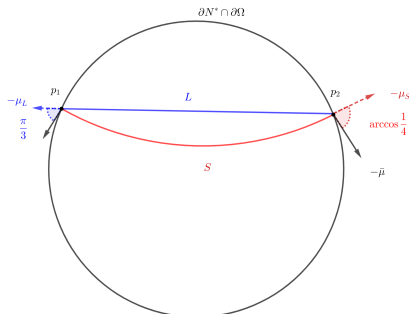
But yes, if (V, W) satisfies a stronger distributional structure.

Main Result: An Example

Let N^* be the unit ball in the plane $\mathbf{R}^2$ centered at the origin, ∂N^* is then the unit sphere. Let p_1, p_2 be two points on ∂N^* such that the line segment joining them, say L , has contact angle $\frac{\pi}{3}$ with ∂N^* . Let S be a mean convex curve joining p_1 and p_2 in N^* to enclose a closed set Ω so that S has a contact angle $\theta_0 = \arccos \frac{1}{4}$ with ∂N^* . Let V be the naturally induced multiplicity 1 varifold by L , let W be the multiplicity 2 varifold induced by $\partial\Omega \cap \partial N^*$.

Then (V, W) is a stationary pair with fixed contact angle θ_0 , such that

$\|V\|, \|W\|$ are supported in Ω , with $p_1 \in \text{spt} V$.

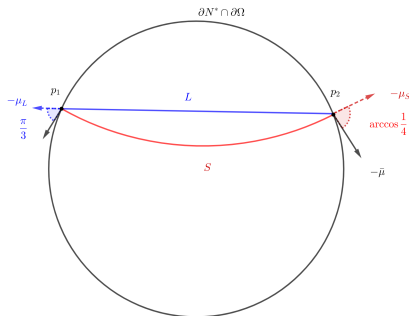


Main Result: An Example

For any $X \in \mathfrak{X}_t(N^*)$,

$$\delta V[X] = \int_L \operatorname{div}_L X d\mathcal{H}^1 = \sum_{i=1}^2 \langle X, -\mu_L(p_i) \rangle = \frac{1}{2} \sum_{i=1}^2 \langle X, -\bar{\mu}(p_i) \rangle$$

since $-\mu_L(p_i) = \cos \frac{\pi}{3}(-\bar{\mu}(p_i)) + \sin \frac{\pi}{3}(-\nu_{N^*}(p_i))$.

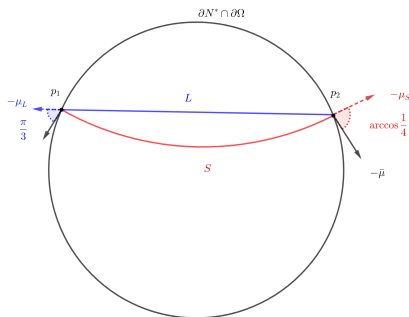


Main Result: An Example

While (recall that $\cos \theta_0 = \frac{1}{4}$)

$$-\cos \theta_0 \delta W[X] = -\cos \theta_0 2 \cdot \int_{\partial N^* \cap \partial \Omega} \operatorname{div}_{\partial N^*} X d\mathcal{H}^1 = -\frac{1}{2} \sum_{i=1}^2 \langle X, -\bar{\mu}_L(p_i) \rangle,$$

thus $(\delta V - \cos \theta_0 \delta W)[X] = 0$ for any $X \in \mathfrak{X}_t(N^*)$.



- This construction works for a general $n \geq 2$.

Main Result

- Testing the distributional structure in the normal direction of the barrier S .

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- Testing the distributional structure in the normal direction of the barrier S .
- Crucial observation:

Theorem (Luigi De Masi ^a)

^a*Rectifiability of the free boundary for varifolds*, Indiana Univ. Math. J. **70** (2021), no.6, 2603–2651.

Let V be a k -varifold on $\overline{U}$ with free boundary at ∂U , then V has bounded first variation with respect to $\mathfrak{X}(U)$. More precisely, there exists a positive Radon measure σ_V on ∂U and a $\|V\|$ -measurable vector field $\tilde{H}$ such that

$$\delta V[X] = - \int_{\overline{U}} \langle X, H + \tilde{H} \rangle d\|V\| - \int_{\partial U} \langle X, \nu_U \rangle d\sigma_V,$$

for any $X \in \mathfrak{X}(U)$, where $\tilde{H}$ is the mean curvature of ∂U if $k = n$ (codim-1).

Main Result

- Recall that for the pair of n -varifolds (V, W) satisfying the contact angle condition θ , $V - \cos \theta W$ has free boundary at ∂U ($\theta \in [\frac{\pi}{2}, \pi)$), it follows immediately

Theorem (Luigi De Masi and Guido De Philippis ^a)

^a*Min-max construction of minimal surfaces with a fixed angle at the boundary*, (2021), arXiv: 2111.09913

Let V be a n -varifold on $\overline{U}$, W be a n -varifold on ∂U , such that (V, W) have contact angle $\theta \in [\frac{\pi}{2}, \pi)$. Then $V - \cos \theta W$ has bounded first variation with respect to $\mathfrak{X}(U)$. More precisely, there exists a Radon measure σ_V on ∂U such that for any $X \in \mathfrak{X}(U)$

$$\begin{aligned} (\delta V - \cos \theta \delta W) [X] &= - \int_{\overline{U}} \langle X, H \rangle d\|V\| \\ &\quad - \int_{\overline{U}} \langle X, \tilde{H} \rangle d(\|V\| - \cos \theta \|W\|) - \int_{\partial U} \langle X, \nu_N \rangle d\sigma_V. \end{aligned}$$

Main Result

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- This formula provides us with a way to test the distributional structure with not only tangential vector field.
- What is the Radon measure σ_V in the smooth case?
- If V is a n -dimensional capillary hypersurface in U and W is the wetting part, then $\sigma_V = \sin \theta \mathcal{H}^{n-1} \llcorner (\partial V)$.
To see this, for any $X \in \mathfrak{X}(U)$,

$$\delta V[X] = - \int_V \langle X, H \rangle d\mathcal{H}^n - \int_{\partial V} \langle X, \mu \rangle d\mathcal{H}^{n-1}$$

where (recall that $\mu = \cos \theta \bar{\mu} + \sin \theta \nu_U$)

$$\int_{\partial V} \langle X, \mu \rangle d\mathcal{H}^{n-1} = \cos \theta \int_{\partial V} \langle X, \bar{\mu} \rangle d\mathcal{H}^{n-1} + \sin \theta \int_{\partial V} \langle X, \nu_U \rangle d\mathcal{H}^{n-1},$$

the first integral on the right relates to $\delta W[X]$, while the last one reveals what σ_V is.

Main Result: fine θ -pair

- Observe also that for any $X \in \mathfrak{X}_t(U)$,

$$\int_W \operatorname{div}_{\partial U} X d\mathcal{H}^n = \int_{\partial W = \partial V} \langle X, -\bar{\mu} \rangle d\mathcal{H}^{n-1} = -\frac{1}{\sin \theta} \int_{\partial U} \langle X, \bar{\mu} \rangle d\sigma_V.$$

Definition (fine θ -pair, I^a)

^aA boundary maximum principle for stationary pairs of varifolds with fixed contact angle, (2022), arXiv: 2205.07643

For $\theta \in [\frac{\pi}{2}, \pi)$, let V be a n -varifold on $\bar{U}$, W be a n -varifold on ∂U , so that (V, W) satisfies the contact angle condition θ . (V, W) is said to be a *fine θ -pair* if there exists $\tilde{\mu} \in L^1(\partial U, \sigma_V)$, with $\tilde{\mu}(x) \in T_x \partial U$ and $|\tilde{\mu}(x)| = 1$ for a.e. $x \in \operatorname{spt}(\sigma_V)$, such that for any $X \in \mathfrak{X}_t(U)$,

$$\delta W[X] = \frac{1}{\sin \theta} \int_{\partial U} \langle X, \tilde{\mu} \rangle d\sigma_V.$$

Main Result: fine θ -pair

Definition (fine θ -pair, II ^a)

^aA boundary maximum principle for stationary pairs of varifolds with fixed contact angle, (2022), arXiv: 2205.07643

For $\theta \in (0, \frac{\pi}{2})$, let V be a n -varifold on $\overline{U}$, W be a n -varifold on ∂U , so that (V, W) satisfies the contact angle condition θ . (V, W) is said to be a *fine θ -pair* if there exists a n -varifold $\tilde{W}$ on ∂U , such that

- ① $(V, \tilde{W})$ is a fine $(\pi - \theta)$ -pair;
- ② For any $X \in \mathfrak{X}_t(\partial U)$,

$$\delta \tilde{W}[X] = -\delta W[X]$$

- $\tilde{W}$ serves as the complement of W . In the smooth case, one can take $\tilde{W} = \partial U \setminus W$.
- This is also true if W is a Caccioppoli set in ∂U .

Main Result: fine θ -pair

- By defining fine θ -pair, when testing the distributional structure, we could transform every term that involves W to a term that involves σ_V . Precisely, for any $X \in \mathfrak{X}(U)$, we have

$$\begin{aligned}\delta V[X] = & \frac{\cos \theta}{\sin \theta} \int_{\partial U} \langle X, -\tilde{\mu} \rangle d\sigma_V - \int_U \langle X, H \rangle d\|V\| \\ & - \int_{\partial U} \tilde{H} \langle X, \nu_{\partial U} \rangle d\|V\| - \int_{\partial U} \langle X, \nu_{\partial U} \rangle d\sigma_V.\end{aligned}$$

- Such a distributional structure leads to a boundary maximum principle.

Main Result: Boundary Maximum Principle

Theorem (Strong Boundary Maximum Principle for Fine θ -stationary pair ^{a)})

^{a)}A boundary maximum principle for stationary pairs of varifolds with fixed contact angle, (2022), arXiv: 2205.07643

Given $\theta_0 \in (0, \pi/2)$, let S be a smooth, compact, properly embedded hypersurface, meeting ∂N^* with a constant angle θ_0 , that is, $\langle \nu_S, \nu_{N^*} \rangle = -\cos \theta_0$ along $\partial N^* \cap S$. Suppose that S is strongly mean convex at a point $p \in \partial S$.

Then, for any fine θ_0 -stationary pair (V, W) , p is not contained in the support of $\|V\|$, if one of the following cases happens:

- i. $\|V\|$ is supported in Ω , $\|W\|$ is supported in T , and $\|V\|(\partial N^*) = 0$;
- ii. $\|V\|$ is supported in Ω , $\|W\|$ is supported in T , and ∂N^* is mean convex in N^* .

Main Result: Sketch of Proof

Lemma (Local capillary foliation near $p \in \partial S \subset \partial N^*$)

For any properly embedded hypersurface S , having constant contact angle $\theta_0 \in (0, \pi/2)$ with ∂N^ , there exists a constant $\delta > 0$; a neighborhood $U \subset N^*$ containing $p \in S \cap \partial N^*$; and foliation $\{S_s\}$ of U with $s \in (-\delta, \delta)$ such that $S_0 = S \cap U$, and each hypersurface S_s meets ∂N^* with constant contact angle θ_0 .*

- Similar with Li-Zhou's argument, we consider the local Fermi coordinate system $(x_1, \dots, x_{n+1})$ centered at p , relative to ∂N^* such that $x_1 = \text{dist}(\cdot, \partial N^*)$, and further assume that $(x_2, \dots, x_{n+1})$ is a local Fermi coordinate system of ∂N^* , such that x_{n+1} is the signed distance function on ∂N^* from ∂S .

Main Result: Sketch of Proof

- Thanks to the contact angle condition of S , we may write S in such a coordinate as the graph $x_{n+1} = f(x_1, \dots, x_n)$.

We could show that the translated graphs of the functions $f_s = f + s$ form a local foliation of N^* near p , each leaf will be denoted by S_s .

This foliation yields a smooth vector field ν on N^* .

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We could show that the translated graphs of the functions $f_s = f + s$ form a local foliation of N^* near p , each leaf will be denoted by S_s .
This foliation yields a smooth vector field ν on N^* .
- By virtue of such a foliation, we define our test vector field $X \in \mathfrak{X}(N^*)$ by

$$X(q) = -\phi(s(q))\nu(q),$$

where

$$\phi(s) = \begin{cases} e^{\frac{1}{s-\epsilon}}, & 0 \leq s < \epsilon, \\ 0, & s \geq \epsilon. \end{cases}$$

Note that $-\nu(q)$ points into N^* for any $q \in \partial N^*$. Such a definition of X agrees with the one in Li-Zhou's proof, up to a different sign.

Main Result: Sketch of Proof

- Testing the distributional structure of the fine θ_0 -stationary pair with X , we obtain

$$\begin{aligned}\delta V[X] = & \frac{\cos \theta_0}{\sin \theta_0} \int_{\partial N^*} \phi(s(q)) \langle -\nu, -\tilde{\mu} \rangle(q) d\sigma_V(q) \\ & - \int_{\partial N^*} \tilde{H}(q) \phi(s(q)) \langle -\nu, \nu_{N^*} \rangle(q) d\|V\|(q) \\ & + \int_{\partial N^*} \phi(s(q)) \langle -\nu, -\nu_{N^*} \rangle(q) d\sigma_V(q).\end{aligned}$$

- Since each leaf S_s meets ∂N^* with constant contact angle θ_0 , so that at every $q \in \partial N^*$, there exists a unit vector $-\bar{\mu}(q) \in T_q \partial N^*$ with

$$-\nu(q) = \sin \theta_0 (-\bar{\mu}(q)) + \cos \theta_0 \nu_{N^*}(q),$$

Main Result: Sketch of Proof

- Combining these facts, we find

$$\begin{aligned}\delta V[X] &= \cos \theta_0 \int_{\partial N^*} \phi(s(q)) \langle \bar{\mu}, \tilde{\mu} \rangle (q) d\sigma_V(q) \\ &\quad - \cos \theta_0 \int_{\partial N^*} \phi(s(q)) d\sigma_V(q) \\ &\quad - \cos \theta_0 \int_{\partial N^*} \tilde{H}(q) \phi(s(q)) d\|V\|(q).\end{aligned}$$

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- Now we invoke the conditions $\theta_0 \in (0, \frac{\pi}{2})$ and that either $\|V\|(\partial N^*) = 0$, or ∂N^* mean convex; also notice that $\langle \bar{\mu}, \tilde{\mu} \rangle \leq 1$, we thus deduce

$$\delta V[X] \leq 0.$$

Main Result: Sketch of Proof

- Finally, if $\text{spt}(\|V\|)$ contains p , then as computed by Li-Zhou, the strongly mean convexity of S at p will then force

$$\delta V[X] > 0,$$

leading to a contradiction.

Main Result: Further Remarks

- The case $\theta_0 \in (\frac{\pi}{2}, \pi)$ cannot be tackled by this approach, unless $\langle \bar{\mu}, \tilde{\mu} \rangle = 1$ a.e. $x \in \text{spt}(\sigma_V)$.

Main Result: Further Remarks

- The case $\theta_0 \in (\frac{\pi}{2}, \pi)$ cannot be tackled by this approach, unless $\langle \bar{\mu}, \tilde{\mu} \rangle = 1$ a.e. $x \in \text{spt}(\sigma_V)$.
- The condition ∂N^* mean convex is not known to be necessary. In other words, we do not have an explicit counterexample showing that the boundary maximum principle shall fail if ∂N^* is not mean convex.

- 1 Contact Angle Condition for Varifolds: Motivation and Definition
- 2 Maximum Principles for Varifolds
- 3 Open Questions

The theory of varifolds with free boundary has been developed both intensively and extensively. Some works that are contextualized in the rich history of the study of this topic:

- Michael Grüter and Jürgen Jost ¹: Allard-type regularity theorem.
- Martin Man-Chun Lee and Xin Zhou ²: (free boundary version of Yau's conjecture) Existence of infinitely many properly embedded minimal hypersurfaces with non-empty free boundary in any smooth compact Riemannian manifold (N^{n+1}, g) for $2 \leq n \leq 6$, with $\text{Ric}(N) \geq 0$ and ∂N convex.

Questions: Capillary version of these results?

¹*Allard type regularity results for varifolds with free boundaries*, Ann. Scuola Norm. Sup. Pisa Cl. Sci. (4) **13** (1986), no.1, 129–169.

²*Min-max theory for free boundary minimal hypersurfaces, I: Regularity theory*, J. Differential Geom. **118** (2021), no.3, 487–553.

Vielen Dank für eure Aufmerksamkeit!